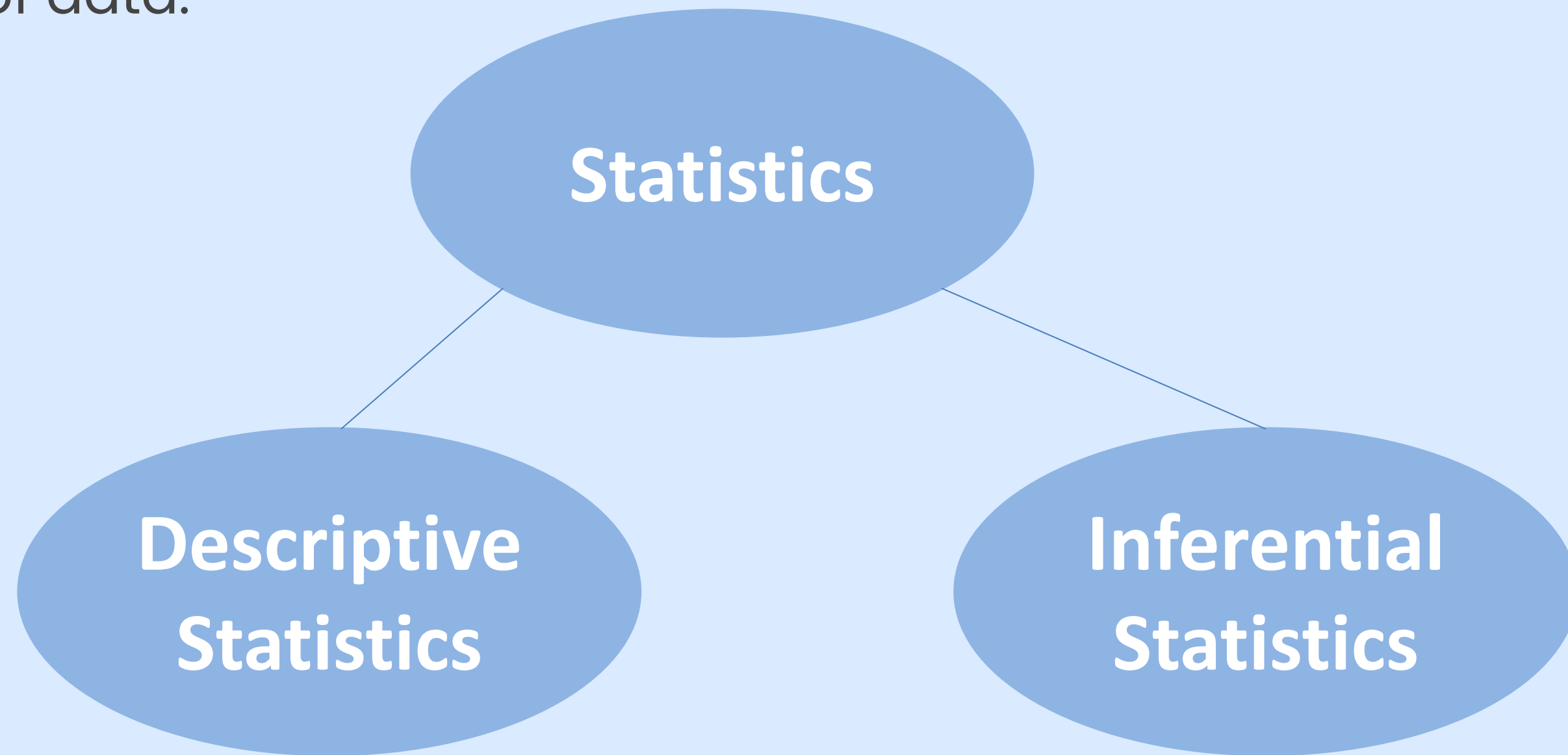
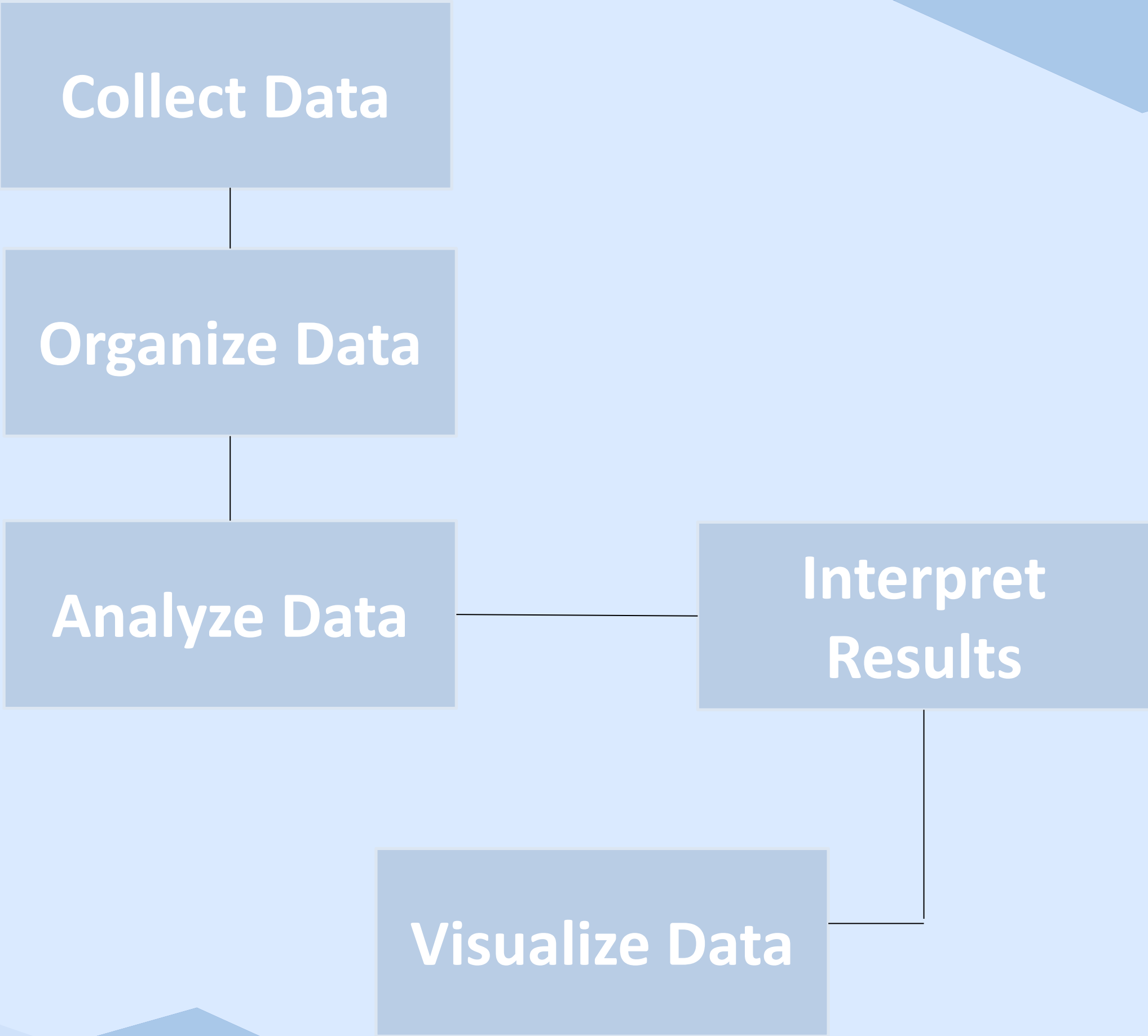


Statistics

Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data.





```
graph TD; A[Collect Data] --> B[Organize Data]; B --> C[Analyze Data]; C --> D[Interpret Results]; D --> E[Visualize Data];
```

Collect Data

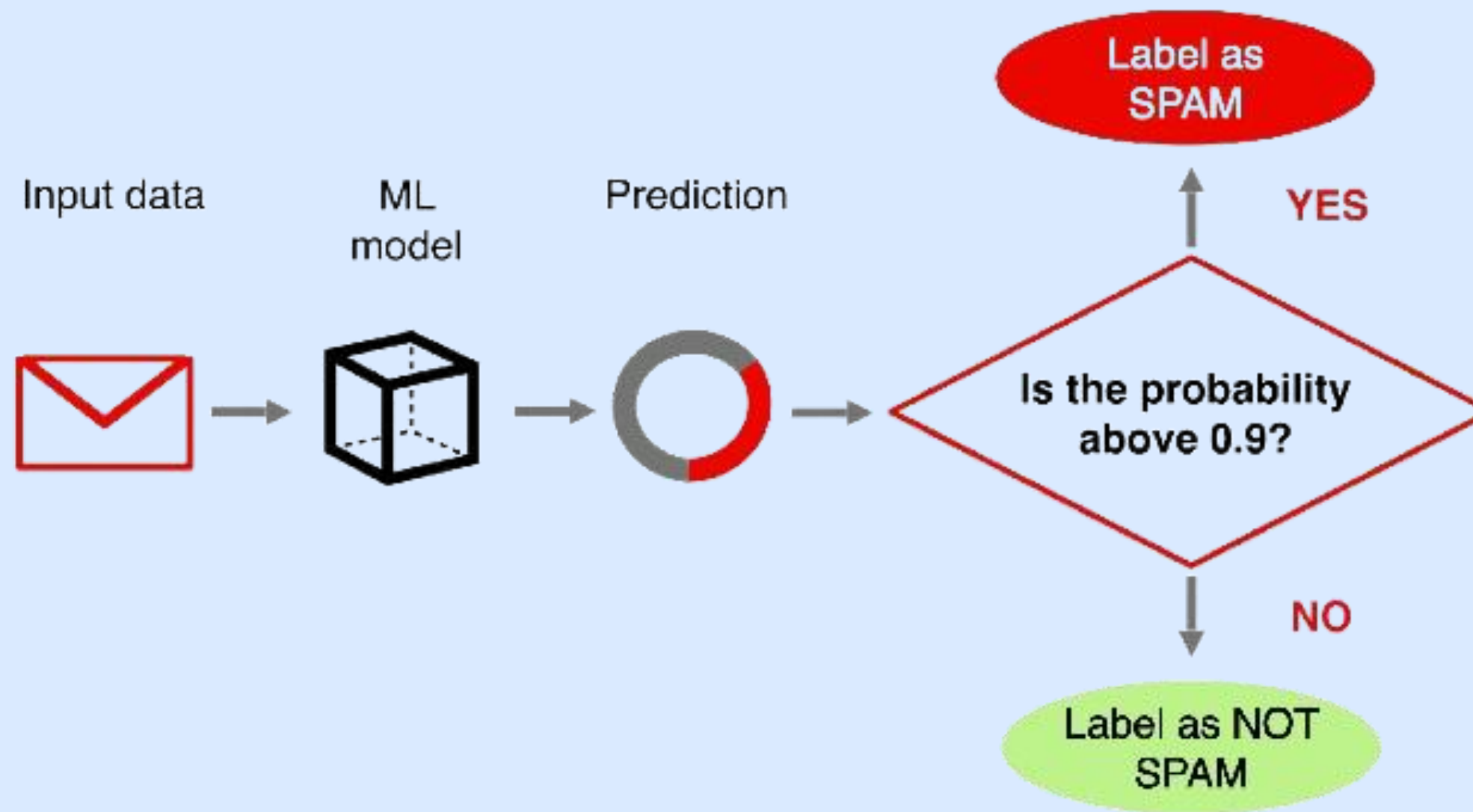
Organize Data

Analyze Data

**Interpret
Results**

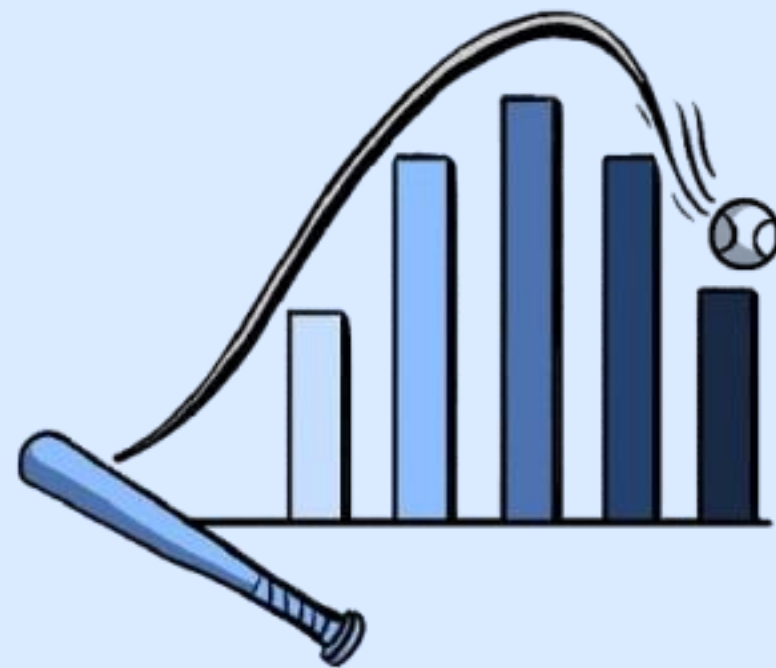
Visualize Data

Example



Descriptive Statistics

Descriptive statistics summarize the characteristics of a data set.



Descriptive Statistics

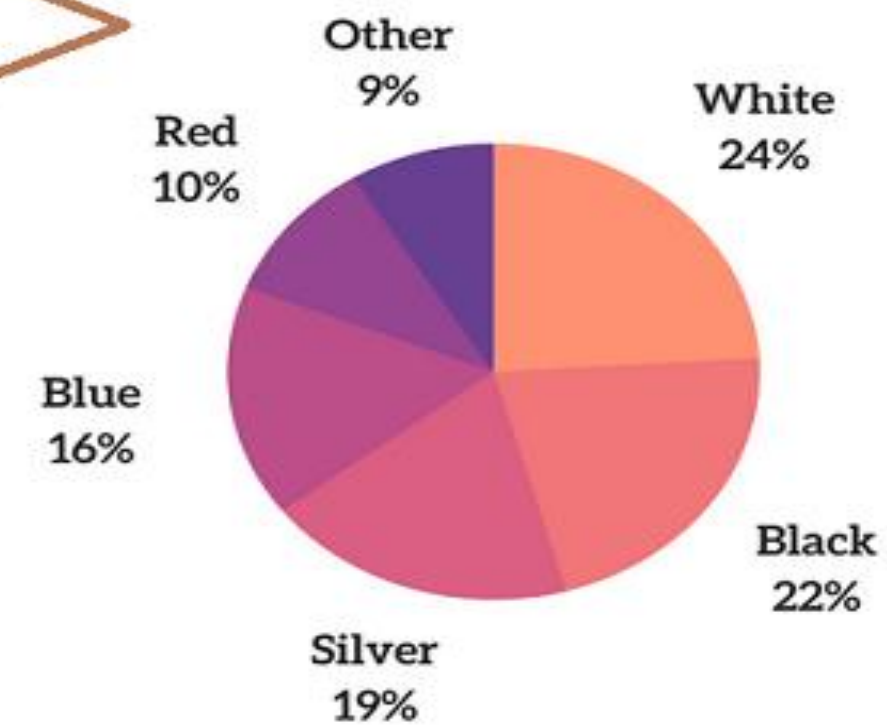
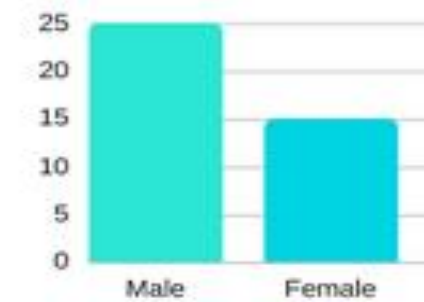
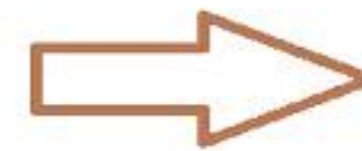
[di-'skrip-tiv stə-'ti-stiks]

Statistics that summarize or describe features of a data set, such as its central tendency or dispersion.

Example

	A	B	C	D
1	Respondent Number	Age	Gender	Favorite Car Color
2	1	22	M	White
3	2	37	F	Silver
4	3	45	F	Black
5	4	62	F	Gray
6	5	28	M	Red
7	6	45	M	Green
8	7	88	F	Brown
9	8	61	M	White
10	9	95	M	Black
11	10	27	M	White
12	11	39	F	Green
13	12	43	M	Brown
14	13	55	F	Black
15	14	59	F	White

RAW DATA

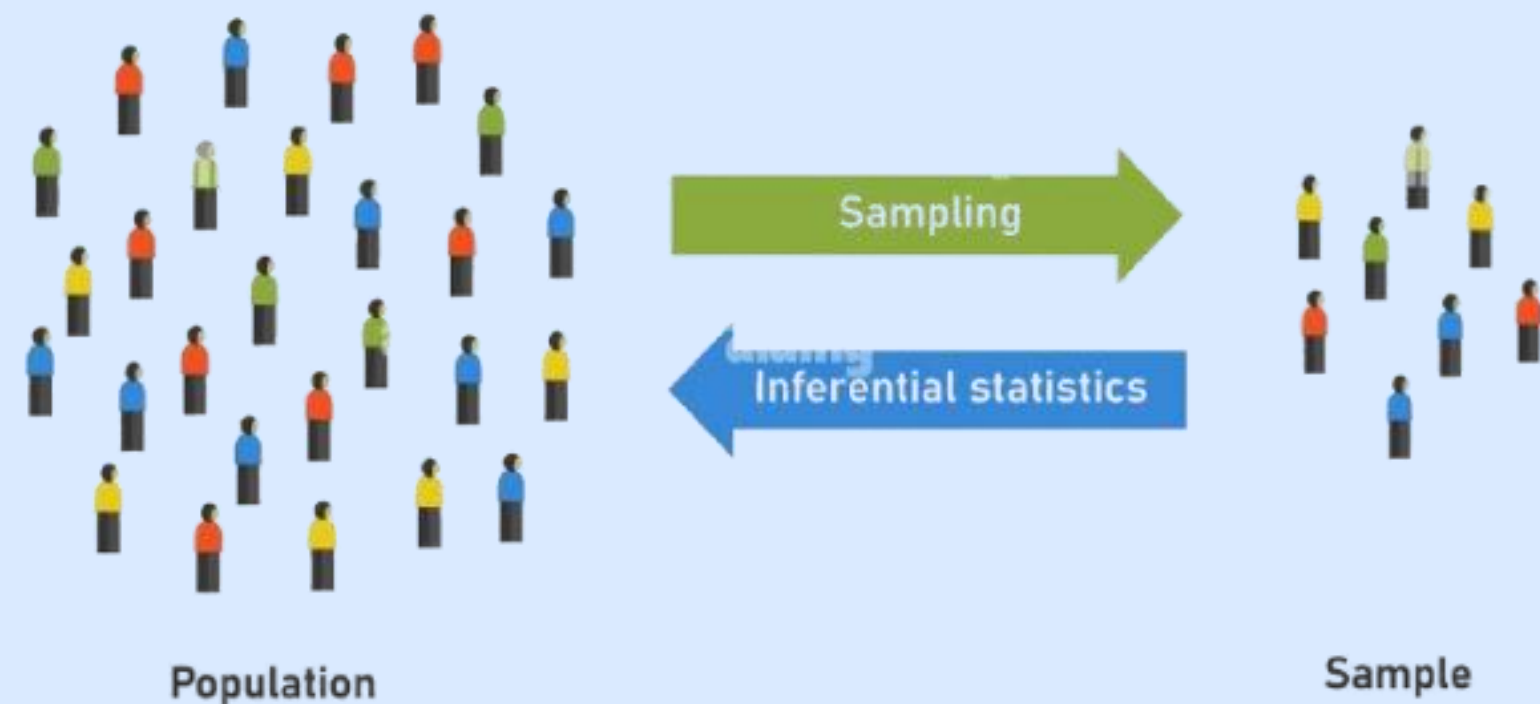


Descriptive Statistics

Inferential Statistics

Inferential statistics is a branch of statistics that **makes the use of various analytical tools** to draw inferences about the population data from sample data

A **population** is the entire group that you want to draw conclusions about. A **sample** is the specific group that you will collect data from.



Census study

- A census study occurs if the entire population is very small or it is reasonable to include the entire population.
- Data obtained from each and every unit of population.
- Results are more representative, accurate and reliable.
- Big problem of underdeveloped countries.



Sampling

- The population of interest is usually too large to attempt to survey all of its members.
- A carefully chosen sample can be used to represent the population.
- The sample reflects the characteristics of the population from which it is drawn.



Sample

['sam-pəl]

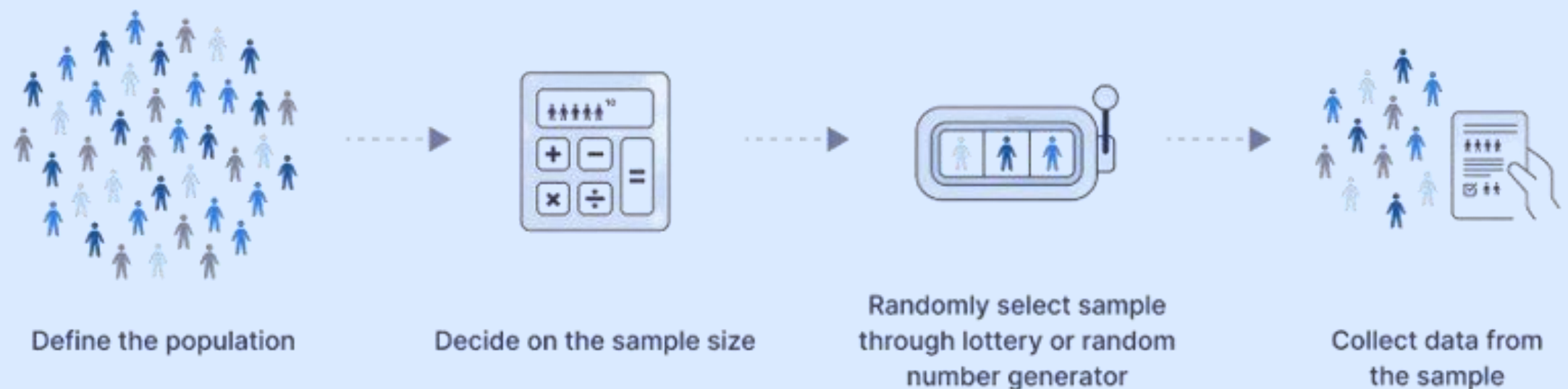
A representative subset
of a population.

Simple random sampling

A type of sampling in which the researcher randomly selects a subset of participants from a population. Each member of the population has an equal chance of being selected.

- Its **quick** to select a sample and provides **randomized** results from a larger pool.
- The sample may not accurately reflect the **target market**.

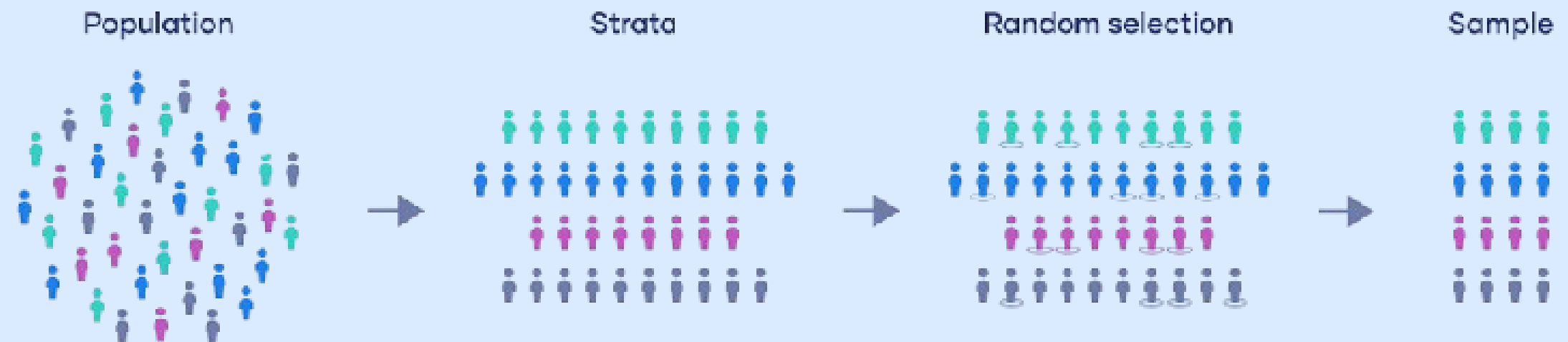
Simple random sampling



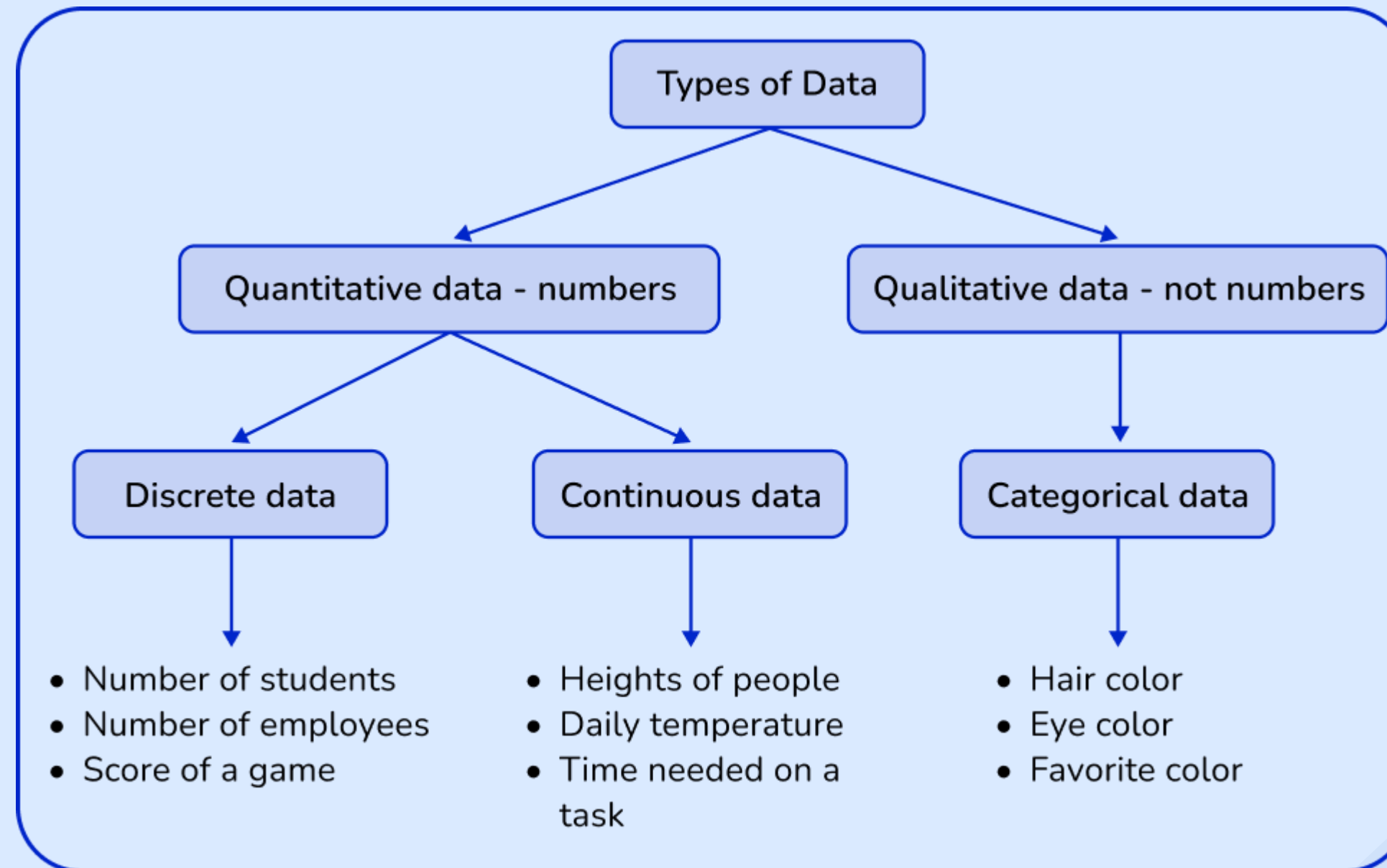
Stratified random sampling

In a **stratified sample**, researchers divide a population into homogeneous subpopulations called *strata* (the plural of *stratum*) or *subgroup* based on specific characteristics (e.g., race, gender identity, location, etc.).

Stratified sampling

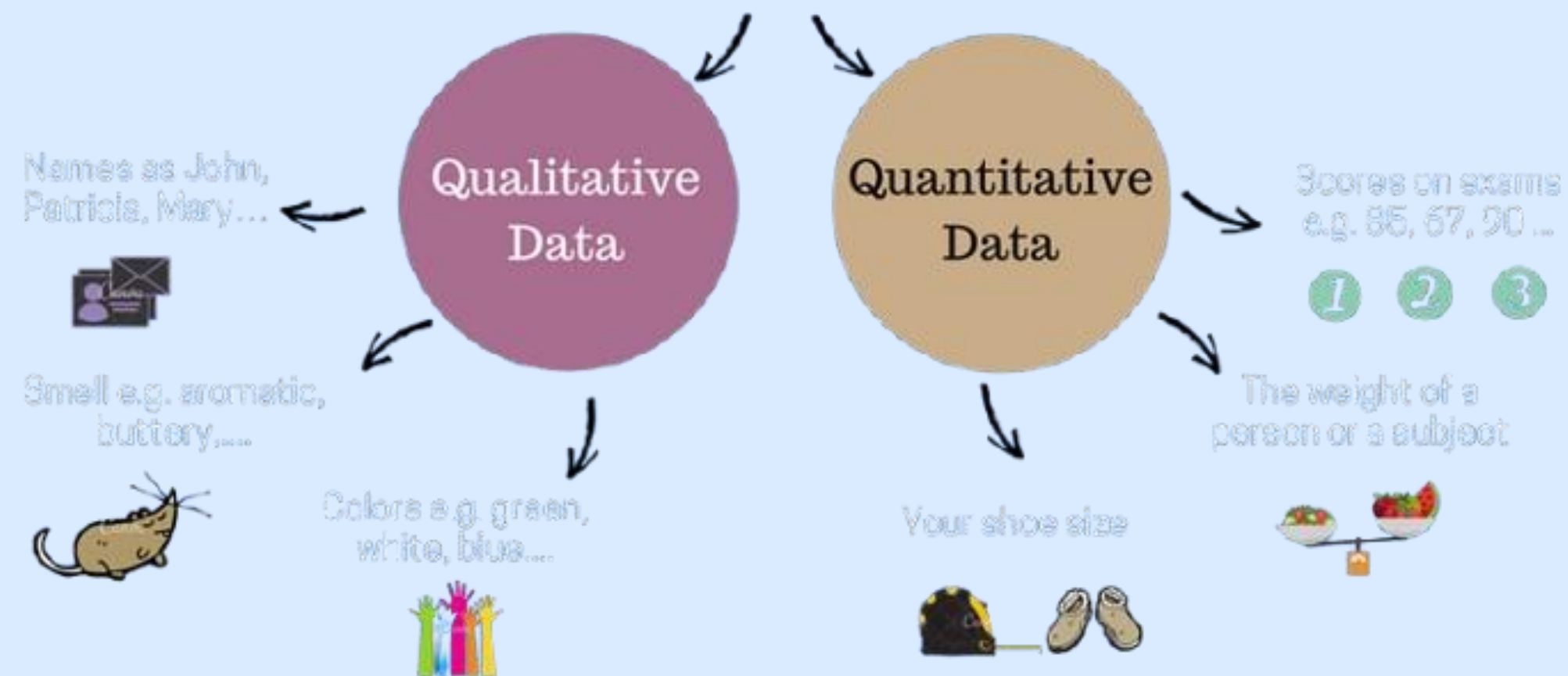


Types of Data

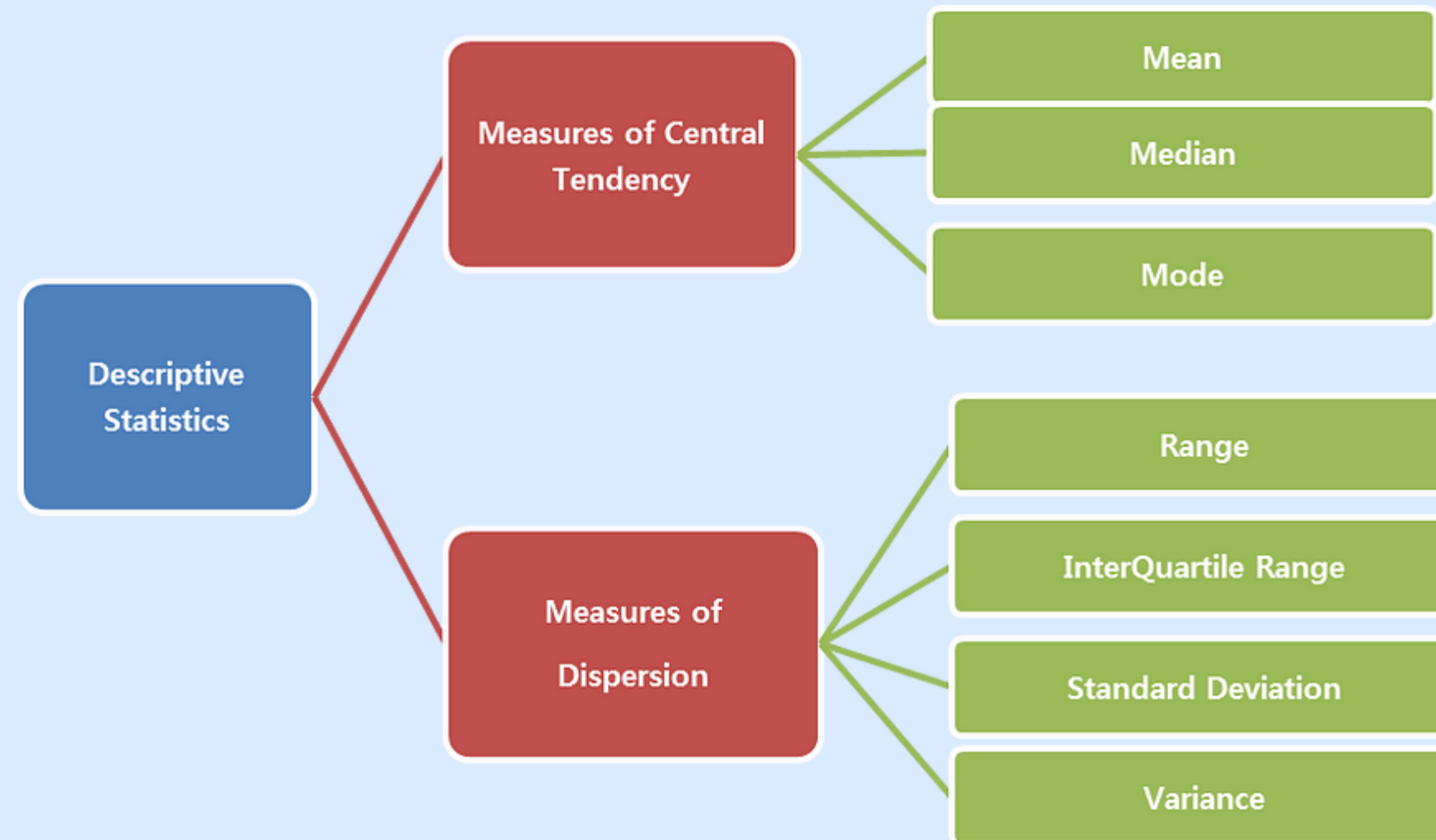


Examples

TYPES OF DATA



Numerical Measures



Measure of central tendency

Measure of central tendency provides us a way to summarize a set of data with a single value that represent the center or middle of the data.

Measure of variability

Variability describes how far apart data points lie from each other and from the center of a distribution.

Simple Mean

The arithmetic mean-or, more succinctly, the mean-is defined as the sum of the observations divided by sample size

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

- The **mean** works well if the data is evenly spread and every value contributes equally.
- Example: Factory Production
A factory produces the following number of items over 5 days: 100, 102, 98, 101, 99.

$$\bar{x} = \frac{100 + 102 + 98 + 101 + 99}{5} = 100$$

It shows the average production rate, helping assess performance.

Sample median

The arithmetic mean-or, more succinctly, the mean-is defined as the sum of the observations divided by sample size

Order the n observations from smallest to largest.

sample median = observation in position $\frac{n+1}{2}$, if n odd.

= average of two observations in positions $\frac{n}{2}$ and $\frac{n+2}{2}$, if n even.

- The median is particularly useful when the data contains **outliers**, as it provides a central value that is not influenced by extreme values.
- **Example:** Income Distribution
Suppose you are analyzing the monthly incomes of 10 employees in a company (in USD):
 - 2,500, 2,800, 3,000, 3,200, 3,500, 3,800, 4,000, 4,500, 5,000, 25,000The data contains an outlier (25,000), which is much higher than the rest of the values. This could distort the mean

Mode

In statistics, the mode is the value that appears most frequently in a dataset. Unlike the mean, which represents the average of the values, and the median, which represents the middle value when the data is ordered, the mode is the value that occurs with the **highest frequency**.

For example, in the dataset {1, 2, 1, 1, 3, 4, 5, 1, 2, 4, 5}; 1 is the most frequent value. Thus, 1 is the mode of this dataset.

- In large datasets, the mode helps find the most frequently occurring value, which can represent a pattern.
Example: Internet Speed Preferences.
A survey asks users to select their preferred internet speed (in Mbps): 10, 20, 10, 30, 20, 10, 10, 30, 20, 10, 50

Mode: 10 Mbps appears 5 times.

Example

Calculation of the sample mean and median

A sample of five university students responded to the question “How much time, in minutes, did you spend on the social network site yesterday?”

100 45 60 130 30

Find the mean and the median.

Solution

The mean is

$$\bar{x} = \frac{100 + 45 + 60 + 130 + 30}{5} = 73 \text{ minutes}$$

and, ordering the data from smallest to largest

30 45 60 100 130

the median is the third largest value, namely, 60 minutes.

The two very large values cause the mean to be much larger than the median. ■

Example

Calculation of the sample median with even sample size

An engineering group receives e-mail requests for technical information from sales and service. The daily numbers of e-mails for six days are

11 9 17 19 4 15

Find the mean and the median.

Solution

The mean is

$$\bar{x} = \frac{11 + 9 + 17 + 19 + 4 + 15}{6} = 12.5 \text{ requests}$$

and, ordering the data from the smallest to largest

4 9 11 15 17 19

the median, the mean of the third and fourth largest values, is 13 requests. ■

Practice Problems

Practice Problems:

What is the mean, median and mode of the following data set?

32, 6, 21, 10, 8, 11, 12, 36, 17, 16, 15, 18, 40, 24, 21, 23, 24, 24, 29, 16, 32, 31, 10, 30, 35, 32, 18, 39, 12, 20

You have the following CPU utilization percentages for a system over 5 minutes: 45%, 50%, 55%, 65%, 70%. What is the average CPU usage during this period?

A survey asks 20 developers about their favorite programming language. The results are: Python (8 votes), JavaScript (6 votes), Java (4 votes), C++ (2 votes). Which programming language is the most popular?

Measures of Variability

Measures of central tendency provide only a partial description of a quantitative data set. The description is incomplete without a measure of the variability, or spread, of the data set. Knowledge of the data set's variability, along with knowledge of its center, can help us visualize the shape of the data set as well as its extreme values. Perhaps the simplest measure of the variability of a quantitative data set is its range

Range

The range of a quantitative data set is equal to the largest measurement minus the smallest measurement

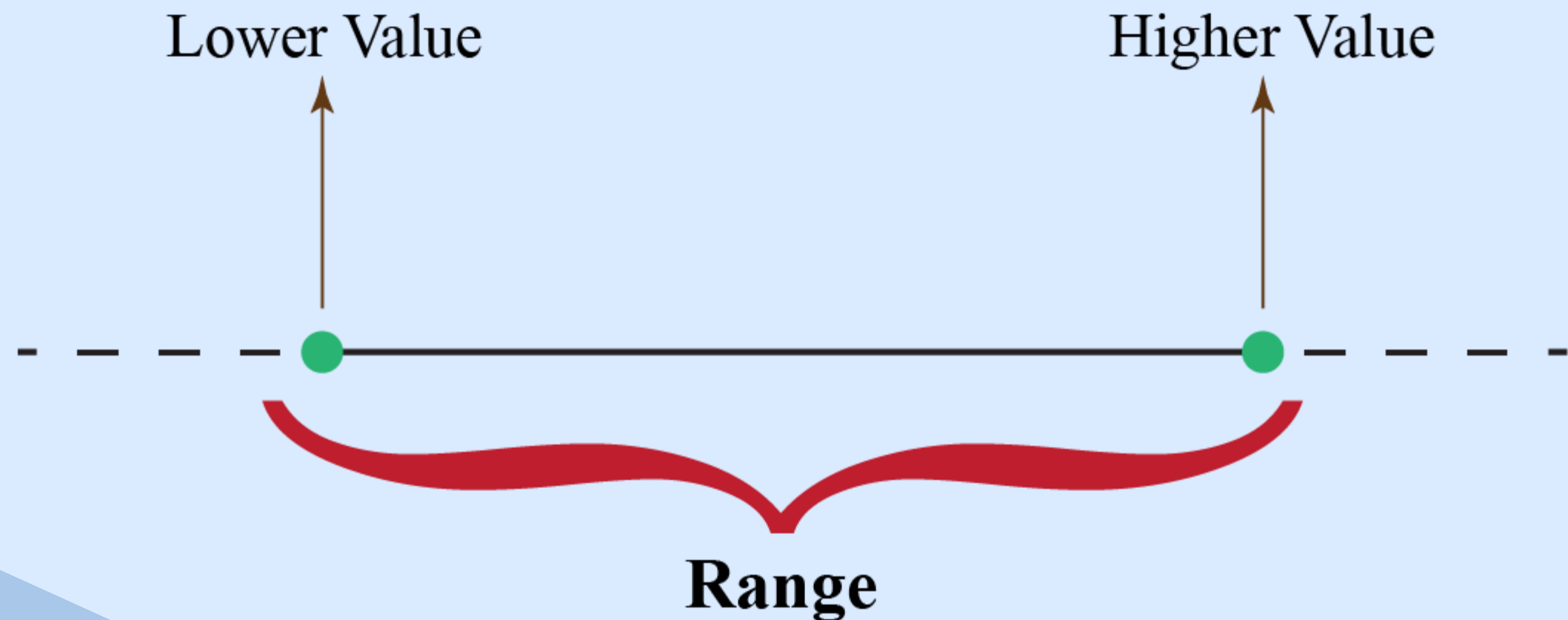
$$\text{Sample Range} = X_{\max} - X_{\min}$$

Example

You measure the network bandwidth (in Mbps) over 5 tests: 50, 60, 55, 100, 45.

Range = $100 - 45 = 65$.

A large range indicates inconsistent bandwidth, possibly due to network congestion.



Variance and Standard Deviation

Definition 1.3: The sample variance, denoted by s^2 , is given by

$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}.$$

The sample standard deviation, denoted by s , is the positive square root of s^2 , that is,

$$s = \sqrt{s^2}.$$

Example

EXAMPLE

Find the variance and standard deviation of the following scores on an exam:

92, 95, 85, 80, 75, 50

SOLUTION

First we find the mean of the data:

$$\text{Mean} = \frac{92+95+85+80+75+50}{6} = \frac{477}{6} = 79.5$$

Then we find the difference between each score and the mean (deviation).

Score	Score - Mean	Difference from mean
92	92 – 79.5	+12.5
95	95 – 79.5	+15.5
85	85 – 79.5	+5.5
80	80 – 79.5	+0.5
75	75 – 79.5	-4.5
50	50 – 79.5	-29.5

Next we square each of these differences and then sum them.

Difference	Difference Squared
+12.5	156.25

Example

+15.5	240.25
+5.5	30.25
+0.5	0.25
-4.5	20.25
-29.5	<u>870.25</u>
Sum of the squares →	1317.50

The sum of the squares is 1317.50.

Next, we find the “mean” of this sum (the variance). $\frac{1317.50}{5} = 263.5$

Finally, we find the square root of this variance. $\sqrt{263.5} \approx 16.2$

So, the standard deviation of the scores is 16.2; the variance is 263.5.

EXAMPLE

Find the standard deviation of the average temperatures recorded over a five-day period last winter:

18, 22, 19, 25, 12

Example

SOLUTION

This time we will use a table for our calculations.

Temp	Temp – mean = deviation	Deviation squared
18	18 – 19.2 = -1.2	1.44
22	22 – 19.2 = 2.8	7.84
19	19 – 19.2 = -0.2	0.04
25 mean	25 – 19.2 = 5.8	33.64
<u>12</u> ↓	12 – 19.2 = -7.2	<u>51.84</u>
96 ÷ 5 = 19.2		94.80 ← sum of squares

To find the variance, we divide 5 – 1 = 4. $\frac{94.8}{4} = 23.7$

Finally, we find the square root of this variance. $\sqrt{23.7} = 4.9$

So the standard deviation for the temperatures recorded is 4.9; the variance is 23.7. Note that the values in the second example were much closer to the mean than those in the first example. This resulted in a smaller standard deviation.

We can write the formula for the standard deviation as $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$

where

- $\sum$ means "the sum of"
- x_i represents each value x in the data
- $\bar{x}$ is the mean of the x_i values
- n is the total of x_i values